

Tata Steel AI Hackathon - Technical Skills Cheat Sheet

1. Artificial Intelligence & Machine Learning

- Local LLM Deployment: Running models locally using Ollama (qwen2.5-coder:7b, mistral) to ensure data privacy.
- Multimodal Computer Vision: Implementing Hugging Face Transformers (Qwen3-VL-8B-Instruct) for image-based defect detection.
- Retrieval-Augmented Generation (RAG): Building an architecture that allows an AI to read PDF documents and answer questions based only on that text.
- Prompt Engineering & Intent Routing: Designing prompts to force the AI to output structured data (JSON) and creating an Intent Router to handle Vision, RAG, and Inventory flows.

2. Backend Engineering

- Python (Advanced): Using advanced Python concepts like Asynchronous programming (asyncio) and multi-threading to prevent the server from freezing during heavy AI tasks.
- FastAPI: Building high-performance, asynchronous REST APIs.
- Server-Sent Events (SSE): Streaming data chunks from the backend to the frontend so the chatbot types out words in real-time.

3. Data Engineering & Databases

- Vector Databases (ChromaDB): Creating, storing, and performing semantic similarity searches on mathematical text vectors.
- Embedding Models: Using Sentence Transformers (BAAI/bge-small-en-v1.5) to convert text into embeddings.
- Relational Databases (SQLite / SQLAlchemy): Designing database schemas for Users, Sessions, Equipment, and Work Orders using an ORM.

4. System Architecture & DevOps Concepts

- Hardware-Aware Engineering: Writing logic to check system memory (RAM/VRAM) and lazy loading massive AI models so the application doesn't crash on weak servers.
- Role-Based Access Control (RBAC): Designing hierarchical workflows separating the permissions of Engineers, Supervisors, and Managers.

How to summarize this if asked in the interview:

"To build this, I acted as a Full-Stack AI Engineer. For the backend, I used Python and FastAPI with SQLAlchemy for database management. For the AI architecture, I built a custom RAG pipeline using ChromaDB and Hugging Face embedding models. Finally, I handled the core AI logic by deploying Ollama and Qwen-VL Vision models locally to guarantee data privacy, and connected them to the frontend using asynchronous Server-Sent Events."